$$C_{\phi q}^{(12)[3]\text{i1\_i2\_}} \to \frac{1}{16\pi^2}\left(\frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,0}[m_2, \tilde{\mu}]\, \delta_{i1i2} - \right.$$
$$\frac{1}{6} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1}[m_2, \tilde{\mu}]\, \delta_{i1i2} + \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] \delta_{i1i2} -$$
$$\frac{1}{6} \tilde{\mu} g_2^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] \delta_{i1i2} + \frac{3}{8} \tilde{\mu} g_2^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] \delta_{i1i2} -$$
$$\frac{1}{2} \tilde{\mu} g_2^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] \delta_{i1i2} + \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] \delta_{i1i2} -$$
$$\frac{1}{8} \tilde{\mu} g_2^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] \delta_{i1i2} - \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} -$$
$$\frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} + \frac{3}{4} \tilde{\mu} g_2^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} -$$
$$\frac{1}{2} \tilde{\mu} g_2^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} + \frac{1}{8} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1}[\tilde{\mu}, m_1]\, \delta_{i1i2} -$$
$$\frac{1}{6} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2}[\tilde{\mu}, m_1]\, \delta_{i1i2} - \frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,1,0}[\tilde{\mu}, m_2]\, \delta_{i1i2} -$$
$$\frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1}[\tilde{\mu}, m_2]\, \delta_{i1i2} + \frac{7}{8} m_2 \tilde{\mu} g_2^4 \text{LF}_{4,1,-1}[\tilde{\mu}, m_2]\, \delta_{i1i2} -$$
$$\frac{1}{2} m_2 \tilde{\mu} g_2^4 \text{LF}_{5,1,-2}[\tilde{\mu}, m_2]\, \delta_{i1i2} + \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] +$$
$$\frac{1}{144} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] -$$
$$\frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] + \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{d}}^{p}, \tilde{\mu}\right] -$$
$$\frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{d}}^{p}, \tilde{\mu}\right] - \frac{1}{144} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{p}\right] -$$
$$\frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] + \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p}$$
$$\text{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] + \frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] -$$
$$\frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i2}\right] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{u}}^{p}, \tilde{\mu}\right] -$$
$$\frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{u}}^{p}, \tilde{\mu}\right] - \frac{5}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{d}}^{p}\right] +$$
$$\frac{1}{48} m_1 \tilde{\mu} g_1^2 \left(-\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \text{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{q}}^{i2}\right] -$$
$$\frac{7}{48} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{u}}^{p}\right] +$$
$$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{d}}^{p}\right] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}\right] \delta_{i1i2} -$$
$$\frac{3}{16} m_2 \tilde{\mu} g_2^2 \left(\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \text{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}\right] +$$
$$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{u}}^{p}\right] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,0}\left[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] -$$
$$\frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1}\left[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{3,1,1,-1}\left[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] +$$
$$\frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1}\left[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{p}\right] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,0}\left[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] +$$
$$\frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1}\left[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{3,1,1,-1}\left[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] -$$
$$\frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1}\left[m_3, m_{\tilde{u}}^{p}, m_{\tilde{q}}^{i2}\right] + \frac{1}{16} \tilde{\mu} \overline{y_d}^{i2p} \overline{y_u}^{rs} y_u^{i1s} a_d^{rp}$$
$$\text{LF}_{2,2,1,-1}\left[m_{\tilde{d}}^{p}, \tilde{\mu}, m_{\tilde{q}}^{r}\right] - \frac{1}{16} \tilde{\mu} \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} a_u^{ps} \text{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{u}}^{s}\right] -$$
$$\frac{1}{16} \tilde{\mu} \overline{y_d}^{i2p} \overline{y_u}^{rs} y_u^{i1s} a_d^{rp} \text{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{r}, \tilde{\mu}, m_{\tilde{d}}^{p}\right] + \frac{1}{16} \tilde{\mu} \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} a_u^{ps}$$
$$\text{LF}_{2,2,1,-1}\left[m_{\tilde{u}}^{s}, \tilde{\mu}, m_{\tilde{q}}^{p}\right] + \frac{1}{24} m_1 \tilde{\mu} g_1^2 \left(\overline{y_d}^{i2p} y_d^{i1p} - \overline{y_u}^{i2p} y_u^{i1p}\right) \text{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}\right] +$$
$$\frac{1}{24} m_1 \tilde{\mu} g_1^2 \left(-\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \text{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}\right] +$$
$$\frac{3}{8} m_2 \tilde{\mu} g_2^2 \left(\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \text{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}\right] -$$
$$\frac{3}{8} m_2 \tilde{\mu} g_2^2 \left(\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}\right) \text{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}\right] - \frac{1}{8} \tilde{\mu} \overline{y_d}^{i2p} \overline{y_u}^{rs} y_u^{i1s}$$
$$a_d^{rp} \text{LF}_{2,1,1,0}\left[\tilde{\mu}, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{r}\right] + \frac{1}{8} \tilde{\mu} \overline{y_d}^{i2p} \overline{y_u}^{rs} y_u^{i1s} a_d^{rp} \text{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{r}\right] -$$
$$\frac{1}{8} \tilde{\mu} \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} a_u^{ps} \text{LF}_{2,1,1,0}\left[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{s}\right] + \frac{1}{8} \tilde{\mu} \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} a_u^{ps}$$
$$\text{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{s}\right] - \frac{1}{48} \tilde{\mu} g_1^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] +$$
$$\frac{1}{96} \tilde{\mu} g_1^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p}$$
$$\text{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] - \frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] +$$
$$\frac{1}{48} \tilde{\mu} g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] - \frac{1}{96} \tilde{\mu} g_1^2 \overline{y_u}^{i2p} a_u^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] + \frac{1}{48} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] +$$
$$\frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] + \frac{1}{16} \tilde{\mu} g_2^2 \overline{y_d}^{i2p} a_d^{i1p}$$
$$\text{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] +$$
$$\frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] + \frac{1}{16} \tilde{\mu} g_2^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] -$$
$$\frac{1}{32} \tilde{\mu} g_2^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] + \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p}$$
$$\text{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}\right] -$$
$$\frac{1}{144} \tilde{\mu} g_1^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}\right] - \frac{1}{96} \tilde{\mu} g_1^2 \overline{y_d}^{i2p} a_d^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] + \frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] +$$
$$\frac{1}{16} \tilde{\mu} g_2^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}\right] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_d}^{i2p} a_d^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] -$$
$$\frac{1}{3} \tilde{\mu} g_3^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}\right] + \frac{1}{96} \tilde{\mu} g_1^2 \overline{y_d}^{i2p} a_d^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}\right] - \frac{1}{96} \tilde{\mu} g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}\right] -$$
$$\frac{1}{32} \tilde{\mu} g_2^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}\right] - \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_u}^{i2p} a_u^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}\right] - \frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}\right] +$$
$$\frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}\right] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}\right] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}\right] -$$
$$\frac{1}{144} \tilde{\mu} g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}\right] + \frac{1}{96} \tilde{\mu} g_1^2 \overline{y_u}^{i2p} a_u^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] +$$
$$\frac{1}{16} \tilde{\mu} g_2^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}\right] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_u}^{i2p} a_u^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}\right] -$$
$$\frac{1}{3} \tilde{\mu} g_3^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}\right] - \frac{1}{96} \tilde{\mu} g_1^2 \overline{y_d}^{i2p} a_d^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}\right] + \frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] +$$
$$\frac{1}{96} \tilde{\mu} g_1^2 \overline{y_u}^{i2p} a_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}\right] - \frac{1}{96} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p}$$
$$\text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_d}^{i2p} a_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}\right] -$$
$$\frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}\right] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_u}^{i2p} a_u^{i1p}$$
$$\left.\text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}\right] - \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}\right]\right)$$